COMMERCIAL ITEM DESCRIPTION

SPECTRUM ANALYZER
10 Hz to 50 KHz

The General Services Administration has authorized the use of this Commercial Item Description.

Scope. This Commercial Item Description (CID), describes a 10 Hz to 50 KHz spectrum analyzer. This CID is meant as a minimum requirement for the spectrum analyzer in which only those manufacturers that meet or surpass the following requirements are qualified per this CID. The spectrum analyzer shall include a tracking generator.

Classification. Equipment covered by this specification may be commercially available equipment modified to the extent necessary to meet the following description. The equipment shall be Type III, class 5, style E and color R in accordance with MIL-T-28800.

Applicable documents.

Specifications and Standards. The following specifications and standards form a part of this CID to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-T-28800	Test Equipment for use with Electrical and Electrical and Electronic Equipment, General Specification for.

(Copies of specifications, standards, publications, and other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity).

AMSC N/A FSC 6625

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Other publications. The following document forms a part of this CID to the extent specified herein. Unless otherwise specified, the issue of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the nongovernment documents which is current on the date of the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI-C62.41-1980

(Application for copies of ANSI publications should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

Order of precedence. In the event of a conflict between the text of this CID and the references cited herein (except for associated detail specifications, specification sheets or MS standards) the text of this CID shall take precedence. Nothing in this CID, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

Salient characteristics. The equipment shall be capable of operation within the accuracies, limits, and specifications herein.

1. Safety. The equipment shall be UL (Underwriters Laboratories) listed and approved and/or shall comply with the type III safety requirements of MIL-T-28800.

a. Leakage current protection. Protection shall be provided from leakage current in excess of 3.5 milliamperes (ma) rms (AC and DC) from any accessible conductive parts of equipment (including control shafts with knobs broken or removed and recessed calibration or adjustment controls) to either pole of the power source for any position of the power switch(es).

b. Radio frequency interference (RFI) capacitors. Any capacitor used in RFI filters shall be rated for at least four(4) times the operating voltage.

c. Type III ventilation openings.

(1) Ventilation openings shall be designed to prevent accidental contact with live electrical circuits and mechanical moving parts.

(2) Top ventilation openings shall incorporate small mesh screening or similar design to prevent the expulsion of hot electronic particles due to failure of electronic devices such as capacitors.

2. Parts, materials, and processes. Except for marking, the spectrum analyzer shall comply with the type III parts, materials, and process requirements of MIL-T-28800. Nonstandard parts approval requests shall be in accordance with MIL-T-28800. Front panel marking shall be in accordance with MIL-T-28800.

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a. Restricted material. The equipment shall comply with the restricted material requirements of MIL-T-28800, including the requirement for mercury and radioactive materials, and shall contain no combination of materials which cause deterioration of any material contained in the equipment due to the effects of outgassing.

3. Design and construction. Unless otherwise specified herein, the equipment shall comply with the type III design and construction requirements of MIL-T-28800. The equipment shall be of solid-state construction. Encapsulation and embedment (potting) of subassemblies may be used only for high voltage circuits where construction is susceptible to leakage or corona, indicators and associated controls shall be in close proximity to each other, all potentiometer controls shall have at least a 270 degree rotation range (4.71 radians).

a. Type of construction. Construction of the spectrum analyzer shall be of a monolithic design.

b. Signal input and output connectors. Unless otherwise specified, all signal input and output connectors shall be type BNC female.

c. Internal cooling. The equipment shall be capable of operating continuously for twenty four(24) hours at 55°C. If forced air is used, it shall be as specified in MIL-T-28800.

(1) Forced air cooling motor. DC brushless motors shall be utilized for the forced air cooling motor in the equipment.

(2) Air filter. Equipment utilizing a cooling scheme which exhausts air from the equipment through the cooling fan shall not require an externally accessible air filter.

(3) Thermal switch. A thermal switch shall be provided which will interrupt circuit power to the spectrum analyzer when the internal temperature rises to the degree of possible damage to the electronic components. A minimum of 10°C hysteresis drop from this tripping temperature must occur before circuit power is again applied.

4. Electrical power sources. The equipment shall operate from nominal commercial, military, and shipboard power sources of 100, 120, 220 and 240 volts ac (±10%) at line frequencies of 50 Hz to 400 Hz.

a. Electrical power connectors. The equipment shall operate with a ground at the primary source on either side of the single phase input.

b. Power consumption. Maximum power consumption of all components combined shall not exceed 100 watts.

c. Fuses. Fuse(s) shall be provided, on the exterior rear panel for replacement without entering the enclosure. The fuse(s) shall automatically disconnect the equipment from the power source in the event of excessive current requirements.

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[redacted]. Surge and transient voltage protection. The spectrum analyzer shall contain surge and transient voltage protection internal of the enclosure. The transient voltage protection circuit shall be so designed as to prevent electronic device damage within the system per type III solid state tolerance bands of MIL-T-28800, the surge voltage protection circuit shall conform to ANSI/IEEE Std. C62.41-1980, 2.5KY, category A. The surge protection circuit shall drain excess power due to surges directly to the ground conductor for personnel protection. Normal operation of the spectrum analyzer may be inhibited and false operational signals may be generated under input voltage transients greater than ±20 percent and frequency transient conditions greater than ±5 percent from within the nominal type III steady-state tolerance bands of MIL-T-28800, damage (other than fuse failure) to the spectrum analyzer shall not occur during voltage transient conditions greater than ±20 percent and frequency transient conditions greater thar ±5 percent within the maximum limits (30 percent voltage transients and 10 percent frequency transients) specified in MIL-T-28800. Normal operation of the spectrum analyzer shall be automatically resumed when the transient has passed and any blown fuses are replaced. Blown fuses shall be externally accessible. Testing shall be in accordance with MIL-T-28800.

5. [redacted]imensions and weight. The size and weight of individual components shall be consistent with current commercial capabilities, refer to paragraphs 5.a and 5.b of this CID.

a. Equipment dimensions. The nominal dimensions per component for the spectrum analyzer should be approximately 13 inches height x 17 inches wide x 22 inches deep.

b. Equipment weight. The weight including all accessories (excluding manual) per component, shall not exceed 40 lbs.

6. E[redacted]nclosure requirements. Unless otherwise specified herein, the equipment [redacted]shall comply with the style E enclosure requirements of MIL-T-28800.

a. Rackmount. Rackmount capability shall be available as an option for the equipment specified herein.

b. Tilt bail handle. A tilt-bail handle shall be provided in lieu of the hinged metal handles in accordance with MIL-T-28800. Exception to this paragraph may be acceptable as this is not a prime consideration of the equipment.

c. Reusable pouch or container. An attached accessory pouch may be used for storage of all supplied accessories in lieu of front-cover storage.

7. M[redacted]arking and identification.

a. Marking. The visible surface adjacent to panel facilities such asvariable controls, indicators, jacks, connectors and fuse holder(s), shall be marked with a suitable word, phrase, or abbreviation thereof, indicating the purpose or use of the part.

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(1) Panel marking. The front panel marking shall comply with the type III parts, materials and processes panel marking requirements of MIL-T-28800; however, marking identification processes other than the type III requirements specified in MIL-T-28800 may be used, provided the marking and identification processes are capable of passing the front panel marking test when tested in accordance with this CID.

8. Environmental requirements.

a. Temperature requirements. Unless otherwise specified herein, the equipment shall meet its specified performance characteristics and other requirements when subjected to the class 5 temperature requirements specified by MIL-T-28800 and as listed below.

(1) Non-operating temperature. The equipment shall meet its performance and accuracy requirements in an non-operating environment of -20°C to +70°C.

(2) Operating temperature. The equipment shall meet its performance and accuracy requirements in an operating environment of 0°C to 40°C.

b. Humidity requirements. The equipment shall meet its required performance and accuracy requirements under humidity requirements for class 5 equipment as specified by MIL-T-28800.

c. Altitude requirements. The equipment shall meet its required performance and accuracy requirements under the altitude requirements for class 5 equipment as specified by MIL-T-28800.

d. Vibration requirements. The equipment shall meet the class 5 vibration requirements as specified by MIL-T-28800.

e. Mechanical shock requirements.

(1) Shock, functional. The equipment shall meet the class 5 shock,pulse requirement as specified by MIL-T-28800.

(2) Transit drop. The equipment shall meet transit drop requirement as specified by MIL-T-28800.

(3) Bench handling. The equipment shall meet the requirements as specified by MIL-T-28800.

(4) Shock, high impact. The equipment shall meet the requirements as specified by MIL-T-28800.

f. Electromagnetic interference (EMI). The equipment shall comply with MIL-T-28800.

g. Magnetic environment. The equipment shall comply with the magnetic environment requirements of MIL-T-28800.

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9. Reliability. Reliability verification testing, shall be in accordance with the type III, class 5 environmental conditions of MIL-T-28800.

a. Mean time between failures (MTBF). The equipment mean time between failures (MTBF) shall be 2,000 hours. The lower test MTBF shall be 2,000 and the upper test shall be MTBF 4,000.

b. Pre-conditioning requirement. Each deliverable equipment shall be capable of being subjected to the environmental screening effects of burn in and thermal cycling, with thermal cycling being accomplished in the sample testing phase.

(1) Burn-in. Burn-in shall expose the equipment to extended periods of elevated temperatures, (40°C for a minimum of 100 hours). Each equipment which fails during burn-in shall be repaired and fully retested, until failure free. Failures which occur during the burn-in test shall be noted and reported but shall not count toward the establishment of MTBF. Prior to and following burn-in, a satisfactory operation test (SOT) shall be performed. A satisfactory operation check (SOC) shall be performed conducted at selected times during the burn-in test.

(2) Thermal cycling. Thermal cycling shall expose the equipment to hot and cold cycles for class 5 equipment per MIL-T-28800. Each equipment which fails during thermal cycling shall be repaired and full cycling test shall be noted and reported but shall not count toward the cycling, a satisfactory operation check (SOC) shall be conducted during the last two upper and lower temperature limits.

10. Calibration and maintenance adjustments. The design of the system shall provide for readily accessible calibration adjustments and maintenance adjustments. These adjustments shall be provided by variable value components which are adjustable by the use of simple means. The calibration by substitution of selected components or parts is unacceptable unless specifically approved. The calibration adjustments, wherever possible, shall be accessible without removal of the instrument case or modules.

a. Maintainability. The equipment shall meet the type III maintainability requirements specified by MIL-T-28800.

11. Interchangeability. The equipment shall meet the type III interchangeability requirements specified by MIL-T-28800.

12. Digital interface. The equipment shall meet the digital interface requirements per MIL-T-28800.

13. Performance characteristics.

a. The spectrum analyzer minimum frequency range shall be 10 Hz to 50 KHz and come with an internal tracking generator.

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b. Controls. Unless otherwise specified herein, built in adjustments and compensating devices shall not be externally accessible. All potentiometer controls shall have at least a 270 degrees rotation range.

(1) Front panel controls. Controls for the following functions shall be located on the front panel.

(a) Frequency span
(b) Sweep time
(c) Resolution bandwidth
(d) On/Off
(e) Reference level
(f) Zero frequency scan
(g) Sweep times
(h) Manual sweep
(i) Frequency
(j) Display storage
(k) Display intensity
(l) Display erase
(m) Log/linear
(n) Vertical positioning
(o) Focus

c. Frequency characteristics.

(1) Frequency range. The instrument shall display signals with frequencies from 10 Hz to 50 KHz.

(2) Frequency dial accuracy. The frequency dial accuracy shall be at least 3 KHz.

(3) Display accuracy. The frequency error between any two displayed frequencies shall be within 4 percent of their indicated separation.

(4) Stability. The frequency stability shall be within 30 Hz per minute and ±5 hz per degree C.

(5) Resolution bandwidth. The bandwidths shall be variable from 10 Hz to 300 Hz with at least two intermediate settings.

(6) Incidental FM. The incidental FM shall be 2 Hz peak to peak or less.

d. Amplitude characteristics.

(1) Dynamic range. The dynamic range shall be at least 80 dB.

(2) Input impedance. The input impedance shall be one megohm unbalanced and 600 ohms balanced.

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(3) Damage level. The maximum input without damage shall be at least 100 volts (DC + peak AC).

(4) Sensitivity. The sensitivity of the instrument is determined by the level of the minimum discernible signal. The minimum discernible signal is defined as a signal level plus the noise level which is two times the noise level. The sensitivity shall be greater than or equal to -120dBm withra 300 Hz resolution bandwidth and 600 ohm input termination.

(5) Reference level. The reference level shall be adjustable in 10 dB steps from -10 dBv to -70 dBv on the log 10 dB per division scale.

(6) Display flatness. The displayed signal level shall not change more than 2 dB with any change in resolution bandwidth or more than 0.3 dB for any frequency span.

(7) Spurious responses. Spurious responses shall be 80 dB below the reference level.

e. Sweep characteristics.

(1) Frequency span. The frequency span shall be from 20 Hz per division to 5 KHz per division with at least two intermediate settings.

(2) Zero frequency scan. The instrument shall have a zero frequency scan control.

(3) Sweep times. The sweep times shall be from 10 millisecond per division to 10 seconds per division.

(4) Manual sweep. Manual sweep shall be provided.

(5) Error indicator. A light emitting error indicator shall be provided to indicate wher the display amplitude is out of calibration due to the sweep speed being too fast in relation to the resolution bandwidth and the dispersion. An error indicator is not required on instruments incorporating an automatic resolution selector.

f. Output characteristics.

(1) Output signals. All vertical and horizontal signals applied to the CRT shall be available from the front or rear panel for external application.

(2) Tracking generator. The instrument shall incorporate a tracking generator to track the instrument tuning from 20 Hz to 50 KHz.

(a) Tracking generator output flatness. The tracking generator output level shall not vary more than 3 $\pm$ 0.5 dB from 100 Hz to 20 KHz.

(b) Tracking generator output impedance. The tracking generator output impedance shall be 600 ohms.

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(c) Tracking generator output level. The tracking generator put level shall be capable of being varied by a ratio of at least 10 to 1. The maximum output level shall be at least 100 millivolts.

g. Display characteristics.

(1) CRT indicator. A CRT shall be used for the display of the in any psignalortion of the specified frequency to which the instrument is tuned. The graticule markings shall include a minimum of eight vertical and ten horizontal major division. The usable surface of the CRT shall indicate a true linear and logarithmic display for the input signal on the vertical scale. Each major vertical division shall be identified for the log -dB scale and linear (voltage) scale. The log scale shall be calibrated linearly in dB and shall cover a minimum of 80 dB on the vertical axis. A camera bezel mount shall be provided to permit screen plates to be taken of the display. Graticule markings shall be internal to the CRT.

(2) CRT type. The CRT shall have display storage capability. The controls available for storage shall be as in a through c or equal.

(a) Store the display.
(b) Erase the display.
(c) Operate in non-store mode.

(3) Vertical display function. A control shall be provided on the front panel to select the linear and logarithmic scale for the vertical deflection on the CRT. In the logarithmic display mode, the CRT shall be calibrated to read absolute power in dBm. In the linear display mode, absolute voltage shall be displayed.

(a) Linear display. The linear display mode shall provide an indication of absolute volts per division with an inaccuracy not greater than ±5 percent of full scale value.

(b) Logarithmic display. The logarithmic display shall provide display of 2 or 4 dB and 10 dB per major division over 80 dB dynamic power range for the 10 dB per division mode at any attenuator setting with an inaccuracy less than ±0.25 dB per dB but not more than ±1.5 dB over the full 80 dB dynamic range.

(4) Intensity. A panel control knob shall be provided to permit adjustment of the average intensity of the trace on the CRT.

(5) Vertical positioning. A vertical positioning control shall be provided to permit vertical centering on the trace on the CRT.

(6) Internal calibration signal. A 10 KHz internal calibration signal shall be available for equipment frequency and amplitude calibration.

h. Connector types. The spectrum analyzer shall have binding post contor necand a BNC type connector. A banana type to BNC type adapter may be included as an accessory.

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14. Calibration interval. After calibration, the equipment shall meet each performance requirement within the tolerances specified for a period of at least 9 months, based on an operating time of 1500 hours.

Preservation, packaging, labeling, and marking. The preservation, packaging, packing, labeling, and marking shall be as specified in the contract or order.

Certification. The contractor shall certify that the product offered meets the salient characteristics of this description and that the product conforms to the producer's own drawings, specifications, standards and quality assurance practices and is the same product offered for sale in the commercial marketplace. Small businesses that cannot offer products with established commercial market acceptability, may compete by submission of bid samples. (The Government reserves the rights to require proof of such conformance before first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

Known acceptable products. The following product(s) are known to have previously met the requirements of this CID and need not resubmit bid samples or certification. All other manufacturers must meet the requirements of this CID and submit to bid testing and certification as required.

Manufacturer	Model	NSN
Hewlett-Packard	3580A-C05-002-E02	

Custodian:
Air Force - 99

Preparing activity
Air Force - 82

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